STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



KPIS

NEAR MISSES: WHAT ALMOST HAPPENED

The foam that fell before Columbia's last flight, how the telling of a near miss changes a decision, seven stages from report to fix, what Heinrich's pyramid does and does not show, what to count, and a card for the event that almost happened.

6 earlier shuttle flights, from 1983 to 2002, on which imagery showed foam breaking off the same ramp of the external tank. None of them ended in disaster.

CAIB, 2003

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Much of what goes wrong at work almost went wrong first. This issue is about the events that ended well by luck: how to notice them, why they so often reassure instead of warn, and what a team does with them before the luck runs out.

The US National Safety Council calls a near miss an event that could have caused harm, where only a fortunate break in the chain of events prevented it. Foam broke off the same part of the shuttle's tank on six flights before the one that destroyed Columbia. In experiments by Catherine Tinsley and colleagues, people who heard of a hazard that had left no harm took more risk. James Phimister and colleagues follow a near miss through seven stages. Heinrich's pyramid of 1931 is still everywhere; its data cannot be checked.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

The foam that fell before

On 16 January 2003, 81.7 seconds after launch, foam broke off the left bipod ramp of Columbia's external tank and struck the left wing. On 1 February the shuttle broke up on re-entry; all seven crew died. The accident board found that the same ramp had shed foam before.

1983

First known foam loss from the ramp. Logged as an in-flight anomaly, then closed.

1990–94

Four more. Only 72 of 113 flights had imagery good enough to tell.

2002

A large loss on STS-112, logged only as an “action” item.

2003

The seventh known loss hits Columbia's wing.

Over 22 years, the Board wrote, foam strikes were normalised into a “maintenance” issue. NASA labelled them “in family”: a problem already experienced, analysed and understood.

OUR READING

A near miss repeated often enough stops looking like a warning and starts looking like proof that all is well.

Source: Columbia Accident Investigation Board, 2003

Seven cases confirmed by imagery, about 10% of the 72 imaged flights. The Board called the pattern a normalisation of deviance.

THE NUMBERS

When luck reassures

Catherine Tinsley, Robin Dillon and Matthew Cronin asked 299 business students whether they would still take a non-refundable cruise that a hurricane had a 30% chance of hitting. Some read that their three earlier cruises had gone without trouble; others also knew someone whose cruise had been hit.

WOULD STILL GO ON THE CRUISE, 2012



How the earlier escape was told mattered more than the reminder. The pattern also held in an evacuation task among 102 emergency managers.

OUR READING

The same event, told as “nothing happened” or as “it almost happened”, leads to opposite decisions.

Source: Catherine H. Tinsley, Robin L. Dillon & Matthew A. Cronin, Management Science 58(9), 2012

Students in a scenario, not staff at work. Delays versus injury made no significant difference; the two kinds of near miss did.

THE MODEL

Seven stages, report to fix

James Phimister and colleagues built their framework (2003) on more than 100 interviews at 20 chemical and pharmaceutical sites. It follows a near miss through seven stages, so that a site can find its weak links.

The seven stages

1. Identification: someone sees that something almost happened
2. Disclosure: it is reported
3. Prioritisation and distribution: how serious, who takes it
4. Causal analysis: why it happened
5. Solution identification: what would stop it
6. Dissemination: those affected hear of it
7. Resolution: the change is made and the case closed

The National Safety Council's 2013 case study adds the conditions: reporting that is non-punitive, anonymous if the reporter wishes, and easy, with each near miss investigated for the weaknesses in the system behind it.

OUR READING

Most programmes count stage 2. Near misses are lost at stage 1, when nobody sees them, and at stage 7, when nobody closes them.

Sources: James R. Phimister, Ulku Oktem, Paul R. Kleindorfer & Howard Kunreuther, Risk Analysis 23(3), 2003 (via Abstract at IDEAS/RePEc; stages as cited by Abdul Majid & Shariff (2020) and Zhang et al. (2023), Process Safety Progress); National Safety Council, 2013

We saw the article only as an abstract; the stage names follow two later papers that cite it. The reading is the editors'.

More in the essay 6S without a poster: order you can see

WHAT THE RESEARCH SAYS

300, 29, 1: what the pyramid shows

In 1931 Herbert W. Heinrich estimated that of 330 accidents, 300 cause no injury, 29 a minor one and 1 a major one. Reviewing all four editions in 2011, Fred Manuele found no data behind the ratio that could be examined.

HEINRICH'S GROUP OF 330 ACCIDENTS, 1931



A “major” injury was any case reported to an insurer. Later tests are mixed. On UK railways, only 3 of 21 types of cause differed between injuries, damage and near misses (Wright & van der Schaaf). In 25,000+ US mines over 13 years, more minor incidents went with a higher chance of a fatal one, but the triangle's shape depended on how severity was classified (Yorio & Moore).

OUR READING

Near misses are worth learning from because they often share causes with accidents. That does not make the ratio a law, or fewer small injuries a promise of fewer deaths.

Sources: H. W. Heinrich, McGraw-Hill, 1931 (via Manuele, 2011; the 1941 edition (Internet Archive)); Fred A. Manuele, Professional Safety 56(10), 2011; Linda Wright & Tjerk van der Schaaf, Journal of Hazardous Materials 111(1-3), 2004 (via Abstract at PubMed); Patrick L. Yorio & Susan M. Moore, Risk Analysis 38(4), 2018

Heinrich is quoted from Manuele and the 1941 edition. Manuele also cites US claims data: small claims fell faster than large ones.

HOW IT IS MEASURED

Count the closing, not only the reports

The NSC treats near-miss reporting as a leading indicator, to be used alongside other measures and reported back to the organisation. It warns against rewarding managers for low recordable injury rates: that kind of incentive has been shown to suppress reporting.

1. **Reports per month**
   Expect the count to rise first: more reports can mean more trust, not more danger.
2. **Share closed**
   Reports with an agreed change carried out, out of all reports.
3. **Days to feedback**
   From the report to the reporter hearing what happens next.
4. **Repeats**
   The same near miss again after its case was closed.

HYPOTHETICAL EXAMPLE, ONE MONTH IN A PARCEL DEPOT

Reports	18, up from 12
Changes	10 agreed, 6 carried out
Feedback	5 of 18 reporters heard back within 7 days

The rising count is good news; the gap is in telling people what their report changed. The numbers are invented.

Sources: National Safety Council, 2013; James R. Phimister, Ulku Oktem, Paul R. Kleindorfer & Howard Kunreuther, Risk Analysis 23(3), 2003 (via Abstract at IDEAS/RePEc; stages as cited by Abdul Majid & Shariff (2020) and Zhang et al. (2023), Process Safety Progress)

The four measures and the example are the editors', after the NSC case study and the stages of Phimister et al.

TOOL OF THE ISSUE

The near-miss card

One card per event, filled in the same day. Write it as something that almost happened, not as something that did not; describe the conditions, not the person.

01 What happened where, when, what you saw; no names

02 What could have happened the worst realistic outcome, and what stopped it

03 Why the conditions behind it; ask why more than once

04 What changes one change that removes the condition, not a reminder to be careful

05 Who follows up, by when one owner and a date

06 Told back when the person who reported heard what changed

Sources: National Safety Council, 2013; James R. Phimister, Ulku Oktem, Paul R. Kleindorfer & Howard Kunreuther, Risk Analysis 23(3), 2003 (via Abstract at IDEAS/RePEc; stages as cited by Abdul Majid & Shariff (2020) and Zhang et al. (2023), Process Safety Progress); Catherine H. Tinsley, Robin L. Dillon & Matthew A. Cronin, Management Science 58(9), 2012

A practice proposed by the editors, after the NSC case study, the seven stages of Phimister et al. and the near misses told as “almost happened” in Tinsley et al.

Open the tool: 5 Whys stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Columbia Accident Investigation Board Report, Volume I
Columbia Accident Investigation Board, 2003
https://s3.amazonaws.com/akamai.netstorage/anon.nasa-global/CAIB/CAIB_lowres_full.pdf

How Near-Miss Events Amplify or Attenuate Risky Decision Making
Catherine H. Tinsley, Robin L. Dillon & Matthew A. Cronin, Management Science 58(9), 2012
https://doi.org/10.1287/mnsc.1120.1517

Near-Miss Incident Management in the Chemical Process Industry
James R. Phimister, Ulku Oktem, Paul R. Kleindorfer & Howard Kunreuther, Risk Analysis 23(3), 2003 · via Abstract at IDEAS/RePEc; stages as cited by Abdul Majid & Shariff (2020) and Zhang et al. (2023), Process Safety Progress
https://doi.org/10.1111/1539-6924.00326

Near Miss Reporting Systems (case study, OSHA and NSC Alliance)
National Safety Council, 2013
https://www.osha.gov/alliances/nsc/nsc

Industrial Accident Prevention: A Scientific Approach
H. W. Heinrich, McGraw-Hill, 1931 · via Manuele, 2011; the 1941 edition (Internet Archive)

Reviewing Heinrich: Dislodging Two Myths From the Practice of Safety
Fred A. Manuele, Professional Safety 56(10), 2011
https://aeasseincludes.assp.org/professionalsafety/pastissues/056/10/052_061_F2Manuele_1011Z.pdf

Accident versus near miss causation: a critical review of the literature, an empirical test in the UK railway domain, and their implications for other sectors
Linda Wright & Tjerk van der Schaaf, Journal of Hazardous Materials 111(1-3), 2004 · via Abstract at PubMed
https://doi.org/10.1016/j.jhazmat.2004.02.049

Examining Factors that Influence the Existence of Heinrich's Safety Triangle Using Site-Specific H&S Data from More than 25,000 Establishments
Patrick L. Yorio & Susan M. Moore, Risk Analysis 38(4), 2018
https://doi.org/10.1111/risa.12869

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



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Every issue, one management question, checked against the best research.

ALL ISSUES

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